QUANTITATIVE METHODS 1

Worksheet 5 – Time series

Exercises

Exercise 1 – Components and models

1. Match long-run trend, recession, recurring Q4 peak and exceptional shock to T_t, C_t, S_t or R_t.
2. Write the full additive and multiplicative models.
3. Explain when multiplicative seasonality is more plausible than additive seasonality.

Exercise 2 – Moving averages and centring

Quarterly demand over two years is

$$80,\ 100,\ 120,\ 140,\ 88,\ 110,\ 132,\ 154.$$

Describe the main pattern, calculate the five 4-quarter moving averages, centre them for periods 3 to 6, and explain why an even-order moving average must be centred.

Exercise 3 – Multiplicative decomposition and forecasting

Year	Q1	Q2	Q3	Q4
2005	82	98	120	150
2006	91	111	137	171
2007	104	126	154	194

Given $\widehat{T}_t = 6.343t + 86.939$ and seasonal coefficients $0.789, 0.903, 1.052, 1.255$, interpret the trend slope, check normalization, forecast the four quarters of 2008, and explain multiplicative deseasonalization.

Exercise 4 – Forecast evaluation

For this exercise, suppose actual 2008 values were $136, 162, 188, 245$. Using the Exercise 3 forecasts, calculate forecast errors, MAE and RMSE, and explain the value of a separate validation period and a naive benchmark.